

Mir Mahathir Mohammad - Curriculum Vitae

Contact

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Education

University of Utah, Salt Lake City, Utah Ph.D. in Computer Science Kahlert School of Computing August 2023 – Present Advisor: Dr. El Kindi Rezig	University of Utah, Salt Lake City, Utah M.S. in Computer Science Kahlert School of Computing August 2023 – December 2025 CGPA: 3.978 / 4.0 Advisor: Dr. El Kindi Rezig	Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh B.S. in Computer Science April 2017 – April 2022 Advisor: Dr. Muhammad Abdullah Adnan
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Research Interests

I focus on data preparation in general and data discovery in particular for data lakes. I built SemDisc, the first end-to-end semantic join discovery system using a query-by-example interface. I also explored how discovered relationships can recommend meaningful ways to organize and sort data. Additionally, I contributed to Buckaroo, a visual data wrangling system that enables users to interactively clean and repair data anomalies through direct manipulation of visualizations. My interests include:

Data Systems • Data Lakes • Data Discovery & Integration • Data Wrangling • Data Cleaning • AI for Data Management

Publications

Publications since starting Ph.D. (2023-Present, ~2.8 years)

- SIGMOD'26:** Mir Mahathir Mohammad, El Kindi Rezig. "Qualitative Join Discovery in Data Lakes using Examples." Accepted at ACM SIGMOD International Conference on Management of Data (SIGMOD'26), 2026. [\[PDF\]](#) [\[Link\]](#) *a system for discovering hybrid join paths (combining semantic and equi-joins) in data lakes using query-by-example, supporting hidden tables and semantic tuple matching*
- VLDB'26 (Demo):** Lu Xing, Mir Mahathir Mohammad, Walid Aref, El Kindi Rezig. "LORY-J: Location-aware Join Discovery from Data Lakes." Accepted at VLDB 2026 (Demo Track). [\[PDF\]](#) *a system for location-aware join discovery in tabular data lakes that treats spatial attributes as first-class signals, automatically composing hybrid join paths (combining equi-, semantic-, and spatial-joins) through an interactive, map-driven interface so users can discover and materialize spatially co-located datasets without prior knowledge of table schemas or join keys.*
- SIGMOD'26 (Demo):** Mir Mahathir Mohammad, El Kindi Rezig. "SemDisc: An End-to-End Query-by-Example Semantic Join Discovery System." Accepted at ACM SIGMOD 2026 (Demo Track). [\[PDF\]](#) [\[Link\]](#) *an interactive demo enabling users to discover hybrid join paths in schema-less data lakes through a query-by-example interface with semantic type annotation and inverted join path indexing*
- SIGMOD'26 (Demo):** Mir Mahathir Mohammad, El Kindi Rezig. "NLDisc: Intent-aware Join Discovery using Natural Language." Accepted at ACM SIGMOD 2026 (Demo Track). [\[PDF\]](#) [\[Link\]](#) *a system for intent-aware join discovery over data lakes using natural language queries with interactive intent disambiguation and join path diversification*
- CIDR'26:** El Kindi Rezig, Mir Mahathir Mohammad, Nicolas Baret, Ricardo Mayerhofer, Andrew McNutt, Paul Rosen. "Towards Scalable Visual Data Wrangling via Direct Manipulation." Accepted at CIDR 2026. [\[PDF\]](#) [\[Link\]](#) *a visual data wrangling system that enables users to clean and repair data anomalies through direct manipulation of interactive visualizations*
- VLDB'25 (Demo):** Akash Khatri, Mir Mahathir Mohammad, El Kindi Rezig. "Sort it Like You Mean It: Discovering Semantically Interesting Attribute Augmentations to Sort Tables." Accepted at VLDB 2025 (Demo Track). [\[PDF\]](#) [\[Link\]](#) *Recommends semantically meaningful ways to sort tables by automatically discovering and augmenting attributes from data lakes using LLMs.*

Undergraduate research

- **IEEE FG'24:** Iftekhar E Mahbub Zeeon, **Mir Mahathir Mohammad**, Muhammad Abdullah Adnan. "BTVSL: A Novel Sentence-Level Annotated Dataset for Bangla Sign Language Translation." Accepted at IEEE FG 2024. *Introduces the first large-scale sentence-level dataset for Bangla Sign Language translation, derived from 60 hours of YouTube news content with professional signers.* [PDF] [Link]
- **Neurocomputing'22:** Md. Ashraful Islam, **Mir Mahathir Mohammad**, Sarkar Snigdha Sarathi Das, Mohammed Eunus Ali. "A survey on deep learning based Point-of-Interest (POI) recommendations." Accepted at Neurocomputing (Journal), 2022. [PDF] [Link] *Categorizes deep learning approaches for POI recommendation systems in location-based social networks.*

Research Experience

University of Utah, Kahlert School of Computing, Salt Lake City, UT

Graduate Research Assistant, August 2023 – Present

Advisor: Dr. El Kindi Rezig

- Developed algorithms for qualitative join discovery in data lakes using example-based queries, enabling efficient dataset integration across heterogeneous tabular data
- Built systems for semantic attribute augmentation and table sorting, improving data discovery workflows for analysts working with complex datasets
- Implemented scalable data wrangling systems with direct manipulation interfaces, handling data transformations efficiently

Bangladesh University of Engineering and Technology, CSE, Dhaka, Bangladesh

Research Assistant, July 2022 – June 2023

Advisor: Dr. Muhammad Abdullah Adnan

- Developed machine learning pipelines for processing and analyzing large-scale video datasets for sign language translation
- Built data collection and annotation systems for creating structured datasets, handling data cleaning

Additional Experience

Everforth, Tokyo, Japan (Remote)

Frontend Developer, April 2022 – June 2023

- Built scalable web applications using Vue.js and CakePHP

Technical Skills

Data Systems & Databases: PostgreSQL, MySQL, MongoDB, Query Optimization, Data Indexing

Data Processing & ML: Pandas, NumPy, PyTorch, TensorFlow, Scikit-learn, Data Wrangling, ETL Pipelines

Cloud & Infrastructure: Docker, Google Cloud Platform, Azure (familiar)

Programming Languages: Python (advanced), JavaScript/TypeScript, C++, Java, SQL

Development Tools: Node.js, Express.js, React.js, Vue.js, Git, Streamlit

Selected Projects

- **Badhan Blood Donation Management System:** Designed and implemented a full-stack blood donation platform with MongoDB backend, serving users across BUET campus with real-time donor matching and request management [GitHub]
- **CNN-Based Object Detection:** Developed deep learning models for real-time object detection using PyTorch[GitHub]
- **Automated Robotic Arm:** Built computer vision and control systems for robotic manipulation tasks using MATLAB, integrating sensor data processing and motion planning algorithms [GitHub]
- **Comparative Analysis of AI Agents for Othello:** Compared 12 Othello AI agents, from heuristic search baselines to reinforcement learning approaches, using a round-robin tournament to analyze their relative performance [GitHub]